Bohemien

Motor Design Workflow

From requirements to validated design ready for manufacture

Overview

Bohemien Motor Designer is a desktop application for designing permanent magnet synchronous motors (PMSM) in the range 1 kW to 1 MW. The workflow moves through three distinct phases: rapid analytical iteration, finite element verification, and manufacturing output. Each phase corresponds to a set of tabs in the GUI application.

The complete workflow is typically completed in 2–4 hours for an experienced engineer: 15–30 minutes for analytical convergence, 30–60 minutes for 2D FEM validation, and 15–20 minutes for the 3D FEM run. The manufacturing output is generated automatically once the design is validated.

Phase	Purpose
Phase 1 — Analytical	Tabs 1–5. Fast iteration: seconds per run. Converge on geometry, winding, and thermal design.
Phase 2 — FEM verification	Tabs 8, 11. Finite element solve: minutes per run. Validate Ke, Ld/Lq, B-field, cogging.
Phase 3 — Output	Tabs 7, 9, 10. Generate DXF drawing, manufacturing specification, mesh visualisation.

Phase 1 — Analytical Design (Tabs 1–5)

The analytical phase is where the initial motor geometry is established. All calculations in this phase use closed-form electromagnetic equations and run in under one second. The typical workflow is to iterate between Tab 1 and Tabs 2–4 until performance, losses, and temperature all meet the requirements.

Phase 1 · Analytical · Tabs 1 – 5 · seconds per run

Step 1

Tab 1 — Design inputs
Poles, slots, bore, airgap, stack length, magnet dimensions, turns per coil, rated voltage and speed

Step 2

Tab 2 — Analytical performance
Back-EMF constant Ke, rated torque, MTPA current angle, field weakening speed, efficiency estimate

Step 3

Tab 3 — Loss calculation
Bertotti iron loss (hysteresis + eddy + anomalous), Dowell copper loss (skin + proximity effect), mechanical loss

Step 4

Tab 4 — Thermal check
Lumped thermal network: winding temperature, tooth temperature, yoke temperature, coolant outlet temperature

Step 5	**Tab 5 — Efficiency map** Colour-coded efficiency across the full speed-torque operating envelope, showing base speed and field weakening region

Key inputs — Tab 1

Parameter	Description
Poles / Slots	Number of rotor poles (e.g. 8) and stator slots (e.g. 48). Determines winding configuration and cogging frequency.
Stator OD / Bore	Outer diameter and bore (air-gap) diameter in mm. Sets the magnetic and structural envelope.
Airgap	Radial clearance between rotor surface and stator bore, typically 0.5–1.5 mm.
Stack length	Axial depth of the active lamination pack in mm.
Magnet thickness	Radial depth of the permanent magnets in mm. Directly sets Ke and the airgap flux density.
Magnet width fraction	Fraction of pole pitch covered by magnet (0 to 1). Typically 0.80–0.87 for SPM.
Turns per coil	Conductor turns per coil side. Sets N_series which determines Ke, Ld/Lq, and Rs.
Rated power / speed	Target output power (W) and rated speed (rpm). Used to compute rated torque and current.

Iteration loop — when to go back to Tab 1

After running Tabs 2–4, check the following. If any condition fails, adjust the relevant parameter in Tab 1 and re-run.

Problem	Adjustment in Tab 1
Winding temp > 130°C	Reduce current density: increase slot area (slot depth) or reduce rated current (increase turns).
Ke off target	Adjust turns per coil (scales Ke linearly) or magnet thickness (scales Ke as Br·tm/(tm+μr·g)).
Efficiency < target	Reduce iron loss: reduce B-yoke by increasing yoke thickness. Reduce copper loss: increase slot fill or wire diameter.
Torque ripple too high	Change pole/slot combination. Common low-ripple combinations: 8p/48s, 10p/12s, 8p/9s.
Field weakening too early	Reduce Ld by shortening airgap or reducing turns. Increases the base speed before flux weakening is needed.

Phase 2 — FEM Verification (Tabs 8 and 11)

Once the analytical design has converged, the finite element method (FEM) solvers provide a higher-accuracy validation. The FEM models the actual geometry of the motor — slot shape, tooth tips, airgap curvature — rather than the simplified analytical approximations.

There are two FEM solvers: a fast 2D solver and a more accurate 3D solver. Run the 2D solver first to check the design is fundamentally correct, then use the 3D solver for the final validation including end-winding effects.

Phase 2 · FEM verification · Tabs 8 and 11 · minutes per run

Step	
6	**Tab 8 — 2D FEM validation** Structured polar mesh, magnetostatic solve, Arkkio torque, flux linkage — gives B-field map, Ke_2d, Ld_2d, Lq_2d, cogging torque sweep
7	**Tab 11 — 3D Nédélec FEM** Tetrahedral mesh, Whitney edge elements, tree-cotree gauge — gives Ke_3d, Ld_3d, Lq_3d including end-winding flux reduction (typically 5–15%)

2D FEM — Tab 8

The 2D FEM solves the scalar magnetic vector potential equation div(v·grad(Az)) = −Jz across the motor cross-section. The mesh is structured polar (no external mesher needed) and runs in 5–30 seconds depending on mesh density.

Control	Guidance
Ang/slot	Angular mesh divisions per slot pitch. 4 = fast/coarse, 8 = good accuracy, 16 = high accuracy. Start with 8.
Radial layers	Mesh layers across the airgap. Minimum 4 recommended. More layers improve torque accuracy.
Rotor positions	Number of angular positions for cogging torque sweep. 12–24 covers one electrical period.

Expected results from 2D FEM:

- Ke_2d should be within 5–10% of Ke_anal (analytical). If more than 20% off, check the turns per coil setting.
- B-field mean in airgap: 0.6–1.0 T for a well-designed SPM. Max at tooth tips: 1.4–2.0 T.
- Cogging torque peak-to-peak should be less than 2% of rated torque for good designs.
- Ld and Lq from 2D FEM should be close to analytical values (within 30%).

3D FEM — Tab 11

The 3D FEM extrudes the 2D polar mesh axially and splits each hexahedral cell into 5 tetrahedra (Freudenthal decomposition). It uses first-order Nédélec (Whitney) edge elements with tree-cotree gauge for uniqueness.

Control	Guidance
Ang/slot	Same as 2D. Use 8 for good accuracy (ratio Ke_3d/Ke_anal should approach 1.0–1.1).
Axial layers	Mesh divisions along stack length. 10 is typical. More layers give better end-winding resolution.
End-winding layers	2–3 layers outside the stack capture the end-winding flux return path.

Expected results from 3D FEM:

- Ke_3d / Ke_anal ratio should be 1.0–1.15 at Ang/slot = 8–12. Ratio > 1.5 indicates the mesh is too coarse.
- Ke_3d is typically 5–15% lower than Ke_2d due to end-winding flux that does not link all turns.
- Ld_3d and Lq_3d are computed using zero-background perturbation for numerical stability.

- Solve time: approximately 400–1300 seconds for a typical 48-slot 8-pole motor at Ang/slot = 8.

Phase 3 — Output (Tabs 7, 9, 10)

Once the FEM validation confirms the design meets requirements, the three output tabs generate the deliverables for the manufacturing and mechanical engineering teams.

Phase 3 · Output · Tabs 7, 9, 10 · seconds to generate

Step	
8	**Tab 9 — Mesh visualisation** Three-panel view: full cross-section with material tags, airgap zoom showing element resolution, radial flux density slice
9	**Tab 10 — DXF export** AutoCAD R12 ASCII DXF: stator OD, bore, slot geometry, rotor, magnets, airgap. All dimensions in mm. Compatible with SolidWorks, AutoCAD, Fusion 360.
10	**Tab 7 — Manufacturing report** 12-section specification: wire gauge, strand count, turns per coil, slot fill factor, winding sequence, insulation class, stack dimensions, magnet grade, housing tolerances

DXF export — Tab 10

The DXF file contains the complete 2D cross-section geometry of the motor. It can be directly imported into mechanical CAD software to begin the structural and housing design. Key entities in the file:

Entity	Description
Stator outline	Outer diameter arc and back-iron boundary.
Slot geometry	All Qs slots with opening, body, and tooth-tip geometry.
Bore circle	Inner stator surface defining the airgap boundary.
Rotor body	Outer rotor iron boundary.
Magnet arcs	Each PM arc with pole extent defined by magnet_width_fraction.
Dimension labels	Key dimensions annotated: OD, bore, airgap, stack (as note), magnet thickness.

Manufacturing report — Tab 7

The report is generated from the validated design parameters and covers everything needed to wind, assemble, and test the motor. It includes:

- Stator lamination dimensions with tolerances
- Wire gauge (AWG and metric) and number of strands in hand
- Turns per coil and total series turns per phase
- Slot fill factor (copper fraction) — target 42–48%
- Winding sequence table: slot number, layer, phase, direction

- Insulation class and temperature rating
- Magnet grade, dimensions, and magnetisation direction
- Rotor dimensions and shaft bore
- Recommended winding resistance (DC) at 20°C
- Test acceptance criteria: no-load back-EMF, winding resistance, insulation resistance

Summary — Complete Workflow at a Glance

Step	Tab
1	Tab 1 — Design inputs
2	Tab 2 — Performance
3	Tab 3 — Losses
4	Tab 4 — Thermal
5	Iterate 1→4
6	Tab 5 — Efficiency map
7	Tab 8 — 2D FEM
8	Tab 11 — 3D FEM
9	Tab 9 — Mesh visualisation
10	Tab 10 — DXF export
11	Tab 7 — Manufacturing report